

COMPARISON OF MAJOR PLANT GROWTH REGULATORS (PGRS)

Feature	Auxins	Gibberellins (GAs)	Cytokinins	Ethylene	Abscisic Acid (ABA)
Nature	Promotory (Weakly Acidic)	Promotory (Acidic)	Promotory (Basic/Purine derivative)	Promotory & Inhibitory (Gaseous)	Inhibitory (Acidic)
Synthesis/ Discovery	First isolated from human urine	Isolated from fungi <i>Gibberella fujikuroi</i>	Kinetin from herring sperm DNA; natural zeatin from corn/coconut	Synthesized by tissues undergoing senescence and ripening fruits	Discovered for its role in regulating abscission and dormancy
Primary Functions	Rooting, Apical Dominance, Flowering	Internode elongation, Breaking dormancy, Seed germination	Cell division (cytokinesis), Nutrient mobilization,	Fruit ripening, Senescence, Abscission, Breaking dormancy	Tolerance to stress, Stomatal closure, Seed dormancy
Senescence	Delays leaf drop (early stages); Promotes in older parts	Delays senescence (extends market period)	Delays leaf senescence via nutrient mobilization	Promotes rapid senescence and abscission	Promotes senescence (associated with aging)
Fruit & Flowering	Flowering in pineapples; Parthenocarpy in tomatoes	Elongates apples; Bolting in beet/cabbage	Promotes new leaves and chloroplasts.	Ripens fruits (respiratory climactic); Flowering in mango/pineapple	NA
Specific Uses	Herbicide (2,4-D); Rooting in cuttings	Malting; Sugar yield (Sugarcane); Early seed (Conifers).	Promoting adventitious shoot formation	Ethephon for fruit thinning (cotton/cherry) ripening	"Stress hormone"; Helps seeds withstand desiccation.
Seed/Dormancy	NA	Breaks dormancy (implied by early seed production)		Breaks seed and bud dormancy (peanut/potato)	Induces dormancy; seed germination prevention.